

Nilesh

Software Engineer · Java · Spring Boot · React · Bengaluru, India

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SUMMARY

Software Engineer with 2+ years at SAP building distributed identity infrastructure (OAuth 2.0, JWT, SAP CDC/Gigya), Kafka-driven pipelines, and AI-assisted experiences (RAG, LLM orchestration) in Java and Spring Boot. Shipped **DocLib** — production React PWA with **60+ active users**. Published open-source npm packages with **200+ downloads**.

TECHNICAL SKILLS

Languages: Java (8, 11), JavaScript, TypeScript, Python **Testing:** JUnit, Mockito, Postman, Newman
Backend: Spring Boot, Spring MVC, Node.js, REST APIs, GraphQL, Microservices
AI / GenAI: RAG Pipelines, LLM Orchestration, Copilot Integration, LangChain, Prompt Engineering
Frontend: React 19, React Router, Vite, Tailwind CSS, PWA (Workbox, IndexedDB)
Distributed Systems: Apache Kafka, Event-Driven Architecture, Eventual Consistency **Databases:** MongoDB, PostgreSQL, Redis
Identity & Security: OAuth 2.0, OIDC, JWT, SAP CDC (Gigya), RBAC, MFA, Zero-Trust
Cloud & DevOps: AWS, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD **Observability:** ELK Stack, Kibana, Grafana

EXPERIENCE

Software Engineer (Scholar) Aug 2024 – Present
SAP Labs · Java, Spring Boot, Kafka, MongoDB, OAuth 2.0, JWT, SAP CDC (Gigya), Kubernetes, AWS Bengaluru, India

- SAP CX Sales Cloud (Jun 2025 – Present)**
- Resolved Kafka consumer lag across Mail Sync and Group Sync topics, **reducing message backlog by 50%** under peak load — chose partition rebalancing over consumer scaling after profiling lag per topic.
 - Architected **event-driven search refresh pipeline** — MongoDB writes fan out through Kafka to distributed search indexes, enforcing eventual consistency across all tenants.
 - Designed **JWT-based zero-trust impersonation system** (Login As Another User) with claim scoping, RBAC validation, and cryptographic audit logging — deployed across enterprise tenants.
 - Built **automated API regression suite** (Postman/Newman, 30+ scenarios) integrated into CI/CD — reduced regression testing from manual execution to **fully automated CI/CD validation**.

- SAP SuccessFactors RCM (Aug 2024 – May 2025)**
- Built **end-to-end candidate onboarding and authentication workflows** using **OAuth 2.0/JWT** with configurable tenant-level policies, rolled out across all enterprise tenants.
 - Integrated **SAP Customer Data Cloud (Gigya)** for phone OTP login, social login, **SMS MFA**, and OIDC/OAuth 2.0 authentication, aligning with **GDPR** and SAP Cloud Security Guidelines.
 - Hardened auth security: disabled Gigya voice code by default, enforced configurable **password policies** (entropy, expiry, history), and resolved phone/email sync edge cases causing login failures.
 - Built **RESTful deduplication APIs** using RFC-compliant email normalization and indexed lookup, eliminating duplicate candidate records across multi-tenant datasets.
 - Diagnosed auth edge cases including **token expiry races**, malformed JWT claims, and invalid sync payloads via defensive validation layers, reducing auth-related production failures.

PROJECTS

DocLib — Offline PWA PDF Reader 2026
React 19, Vite, Dexie.js, Workbox, IndexedDB, GenAI (Groq / Gemini / OpenAI) doclib.pro [GitHub](https://github.com)

- Shipped production PWA (iOS, Android, desktop) with **60+ active users** — full offline support via Workbox service workers, versioned IndexedDB schema with live migrations.
- Built multi-provider **Ask AI** (Groq/Gemini/OpenAI) with BYOK, AES-GCM encrypted key storage, and RAG-style PDF context injection; implemented prompt injection defenses — all client-side, zero backend.
- Deployed on Netlify with CI/CD on git push; React Router SPA, annotation engine, reading stats dashboard, and AI flashcard generation.

VendorIQ.AI — Invoice Automation Platform 2025
Node.js, Python, Apache Kafka, MongoDB, LangChain, RAG, Google APIs [GitHub](https://github.com)

- Automated end-to-end invoice ingestion from Gmail using a **3-service Kafka pipeline**, eliminating manual processing.
- Implemented **RAG pipeline** over invoice data enabling natural language querying; integrated Gmail, Drive, and Sheets APIs for automated reporting.

OPEN SOURCE & PUBLISHED PACKAGES

deadbutton — Dead UI Element Detector

TypeScript, npm · 200+ downloads

2024
npm npm Live

- npm package detecting unused UI elements via click telemetry; red/yellow/green overlays with **confidence-scored deletion recommendations** — zero backend, zero privacy exposure.
- Implemented **stateless confidence scoring algorithm** factoring element age, click frequency, and recency decay; usable via CDN one-liner or npm import.

@nileshkr17/generateusername — generates 5 username variants from name + DOB

JavaScript, npm

2022
npm npm

EDUCATION

BITS Pilani WILP

M.Tech in Software Engineering

2024 – 2026

Silicon Institute of Technology, Bhubaneswar

B.Tech in Computer Science & Engineering

2020 – 2024